

1. Find the number of edges in the graphs

1. K_4

Relatively simply, we observe that since every pair of distinct vertices is joined by an edge, the number of edges in K_n is

$$\binom{n}{2} = \frac{n(n-1)}{2}.$$

Therefore, we have:

$$|E(K_4)| = \binom{4}{2} = 6.$$

2. K_5

Similarly,

$$|E(K_5)| = \binom{5}{2} = 10.$$

3. K_n , for any positive integer $n \geq 2$

We find that in the complete graph K_n , each edge corresponds to a choice of 2 distinct vertices from the n vertices. Therefore

$$|E(K_n)| = \binom{n}{2} = \frac{n(n-1)}{2}.$$

2. Determine whether the graphs in the figure are isomorphic.

We find the two graphs are isomorphic.

First, we observe that both graphs have the same degree sequence:

$$(3, 3, 3, 3, 2, 2, 2, 2).$$

However, we must perform another check: look at the vertices of degree 2.

For the left graph, the degree-2 vertices are

$$c, d, e, f.$$

Among these vertices, there are *no* edges joining one degree-2 vertex to another.

For the right graph, the degree-2 vertices are

$$s, t, u, v.$$

But here we do have edges between degree-2 vertices, namely

$$su \quad \text{and} \quad tv.$$

We realize that an isomorphism must preserve vertex degrees and adjacency relations. Therefore, if the two graphs were isomorphic, the pattern of adjacency among degree-2 vertices would have to be the same in both graphs. It is not.

Therefore, we see the graphs are not isomorphic.

3. Find the number of distinct labeled path graphs on n vertices.

We find that a path graph P_n has n vertices arranged in a line. If we label the vertices with $1, 2, \dots, n$, there are $n!$ ways to assign the labels to the vertices.

However, reversing the path gives the same labeled graph structure. Trivially, each labeled path is counted twice among the $n!$ orderings:

$$(v_1, v_2, \dots, v_n) \quad \text{and} \quad (v_n, v_{n-1}, \dots, v_1)$$

determine the same path graph.

For $n \geq 2$, the only automorphisms of P_n are the identity and reversal, so each distinct labeled path is counted exactly twice.

Therefore, the number of distinct labeled path graphs on n vertices is

$$\frac{n!}{2}.$$

4. Let $G = (V, E)$ be a 4-regular graph with $|V| = 10$.

1. Show that $\kappa(G) = 1$ or 2 .

We find that $\kappa(G)$ denotes the number of connected components of G .

Since G is 4-regular, every vertex has degree 4. Therefore, in any connected component of G , every vertex must have 4 neighbors inside that same component. So each connected component must contain at least 5 vertices.

Because $|V| = 10$, the graph can have at most two connected components: if it had three or more components, then it would need at least

$$5 + 5 + 5 = 15$$

vertices, which is impossible.

Also, every graph has at least one connected component. Therefore, we find

$$\kappa(G) \in \{1, 2\}.$$

So

$$\kappa(G) = 1 \quad \text{or} \quad \kappa(G) = 2.$$

2. Find examples that $\kappa(G) = 1$ and $\kappa(G) = 2$.

Example with $\kappa(G) = 2$:

Take

$$G = K_5 \cup K_5.$$

Each copy of K_5 is 4-regular, so their disjoint union is also 4-regular on 10 vertices. It has exactly two connected components, so

$$\kappa(G) = 2.$$

Example with $\kappa(G) = 1$:

Let the vertex set be

$$V = \{0, 1, 2, \dots, 9\},$$

and join each vertex i to

$$i \pm 1 \pmod{10} \quad \text{and} \quad i \pm 2 \pmod{10}.$$

Then each vertex has degree 4, so this is a 4-regular graph on 10 vertices. It is connected, since the edges $i \sim i \pm 1$ already contain a 10-cycle. Therefore

$$\kappa(G) = 1.$$